Blockchain-Enabled Federated Learning for Enhanced Collaborative Intrusion Detection in Vehicular Edge Computing

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***Abstract*— Intelligent Transportation Systems (ITSs) are transforming the global monitoring of road safety. These systems, including vehicular networks and transportation infrastructure, are vulnerable to several security issues, which could disrupt services and potentially cause harm to the users. It is crucial to establish robust security measures to protect against evolving attacks and ensure the safe and reliable operation of ITS. Artificial Intelligence (AI)-based Intrusion Detection Systems (IDS) are mainly used to enhance the security of ITS. The adoption of AI-based techniques to secure ITS against new emerging threats has been limited due to a lack of realistic and recent data on these types of attacks (*i.e.*, zero-day attacks). In this context, we introduce a novel Edge-based Framework that uses Federated Learning (FL) and blockchain to secure ITS against new emerging threats. In particular, our proposed framework consists of (1) a novel distributed Edge-based architecture that allows multiple Edge nodes to securely collaborate while preserving their privacy; and (2) a decentralized and secure reputation system based on blockchain technology to maintain the reliability and trustworthiness of the FL process within the ITS; This system manages reputation data for individual nodes (such as vehicles), guaranteeing the integrity of the FL training process. Experiment results using the UNSW-NB15 dataset show that our proposed framework achieves high accuracy and F1 score (99%) in detecting new threats while ensuring the privacy and reliability of the whole ITS. These results demonstrate the effectiveness of our proposed framework in securing ITS.**

***Index Terms*— Sustainable intelligent transportation systems, green vehicular networks, intrusion detection systems, edge computing, federated learning(FL), blockchain.**

I. Introduction

INTELLIGENT Transportation Systems (ITS) are a form of transportation management that utilizes new emergent technologies to improve the efficiency, safety, and sustainability of the current transportation systems. These systems include a variety of technologies, such as traffic signal synchronization, Electronic toll collection (ETC), and real-time public transportation information. These systems, including vehicular networks and transportation infrastructure, are complex and require careful management to ensure their efficient operation. Vehicular networks, also known as Vehicle-to-Everything (V2X) communication, refer to the communication technologies used to connect vehicles with other vehicles, infrastructure, and pedestrians. These technologies enable vehicles to exchange information, such as location, speed, and traffic conditions, in real-time. Both vehicular networks and ITS have the potential to greatly improve transportation, but they also raise security and privacy concerns. Additionally, the collection and storage of sensitive information, such as location data, by ITS, may raise privacy concerns for individuals. It is crucial to establish robust security measures to protect against evolving attacks and ensure the safe and reliable operation of ITS. To address these security and privacy concerns while still promoting sustainability, it is important to prioritize privacy-aware secure collaborative learning methods.

Artificial intelligence (AI)-based intrusion detection systems (IDS) are mainly investigated to enhance the security of ITS. IDSs are largely used to enhance the security of ITS. An IDS is a type of security software that monitors network traffic and identifies potential threats or unauthorized activity. There are two main types of IDS: (1) Network-based IDS which monitors network traffic and looks for patterns or anomalies that may indicate a security threat; and (2) Host-based IDS, on the other hand, which monitors the activity on a specific computer or device and looks for signs of an intrusion. In the context of sustainable ITS, an IDS can be used to detect and prevent cyber attacks on the network or specific devices within the ITS. For example, an IDS could detect and prevent a hacker from accessing traffic control systems or altering traffic patterns. In addition to an IDS, other security measures can be implemented to protect sustainable ITS, such as encryption, regular software updates, and access controls. By implementing a combination of these measures, we can ensure the safe and secure operation of sustainable ITS solutions.

The adoption of AI-based techniques to secure ITS against emerging threats has been limited due to a lack of realistic and recent data on these types of attacks. Without this data, it is

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difficult for AI algorithms to accurately identify and mitigate new threats (*i.e.*, zero-day attacks) as they arise. To build robust AI-based models for ITS infrastructure, it is essential for organizations to collaborate in a privacy-aware manner. With the increasing amount of personal data being collected and processed by ITS, organizations need to prioritize the privacy of individuals and ensure that their personal information is being handled responsibly [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12].

Federated Learning (FL) is a distributed machine learning approach in which multiple parties collaborate to build a model without exchanging their sensitive data. Instead of sending data to a central server to train a model, each party/agent trains a model locally on its own data and sends only the model parameters to the server. The server then combines the model parameters from all parties to produce a global model. One potential application of FL in ITS is to improve traffic prediction and control. For example, traffic data from multiple sources, such as sensors on roads, traffic cameras, and GPS data from vehicles, could be used to train a global model for traffic prediction. The model could then be used to optimize traffic flow and reduce congestion. Blockchain technology could potentially be used to secure ITS by providing a tamper-proof record of transactions and data. For example, blockchain could be used to track the movement of vehicles and ensure that only authorized vehicles have access to certain areas. Blockchain can also be used to store data related to maintenance and repairs, helping to ensure that vehicles are properly maintained and safe to operate.

In this context, we propose a novel Edge-Enabled Framework that uses Federated Learning and blockchain to secure the ITS network against emerging threats (*i.e.*, zero-day attacks). The use of Edge computing, blockchain, and FL can enhance the security and privacy of ITS. Edge computing brings security mechanisms closer to end devices, reducing delays in communication and improving the overall security of the system. FL, on the other hand, enables privacy-aware collaborative learning between vehicles while preserving their privacy. While Blockchain ensures the trustworthiness and reliability of the FL training process. By leveraging these technologies, we build secure and privacy-aware ITS-based models that benefit both individuals and the overall transportation network. In particular, our proposed ITS-based framework consists of (1) a novel distributed Edge-based architecture that allows multiple Edge nodes to securely collaborate to secure the ITS while preserving their privacy; and (2) a blockchain-based reputation system that ensures the trustworthiness and reliability of the FL training process in ITS by managing the reputation data for each node (*e.g.*, vehicle) participating in the learning process in a decentralized and secure manner. The UNSW-NB15 dataset, which includes well-known attacks and malware, was used to conduct in-depth experiments to evaluate the effectiveness of decentralized machine learning models in detecting these threats. The results of these experiments showed a significant accuracy (99%) and detection rate (99%), outperforming centralized machine learning and deep learning models. In addition to their superior performance, our proposed framework also significantly reduced delays and preserved privacy. These results demonstrate the potential of decentralized learning to enhance the security of ITS and other IoT systems, while also protecting the privacy of individuals (*e.g.*, vehicle).

The main contributions of this paper are summarized as follows:

- We design a novel distributed Edge-based architecture that allows multiple Edge nodes to securely collaborate to secure ITS while preserving their privacy. This architecture allows Edge nodes to securely collaborate and share information and knowledge, ensuring the integrity and reliability of the ITS while preserving the privacy of each node.
- We propose a decentralized and secure reputation system based on blockchain technology to maintain the reliability and trustworthiness of the FL process within the ITS. This system manages reputation data for individual nodes (such as vehicles), guaranteeing the integrity of the FL training process.
- We evaluate our proposed framework in terms of accuracy, F1-score, precision, detection rate, and computational complexity. The UNSW-NB15 dataset, which includes well-known attacks and malware, was used to conduct in-depth experiments to evaluate the effectiveness of our proposed framework in detecting zero-day attacks. The obtained results show a significant accuracy (99%) and detection rate (99%), outperforming centralized machine learning and deep learning models. In addition to their superior performance, our proposed framework demonstrates its ability to effectively achieve trustworthiness, efficiency, security, and privacy.

This paper is organized as follows. Section II presents related work. Section III presents our system model. Section IV discusses our blockchain-based reputation system. Section V presents the experimental setup and results. Section VI evaluates the performance of our proposed framework and compares it with recent AI schemes. Finally, Section VII concludes the paper and outlines future work.

II. Related Work

AI techniques have been largely discussed in the literature to detect intrusions in vehicular systems. One approach is to use supervised learning algorithms to classify normal and abnormal behavior in the vehicle's system. In this context, Singh et al. [13] proposed an AI model that can be used to identify false position information transmitted by misbehaving vehicles. The model was constructed using two different classifiers, i.e., logistic regression and support vector machine. Ghaled et al. [14] a new method for detecting security threats in Vehicular networks by combining shared knowledge and Ensemble Learning (EL) techniques. They proposed generating weighted random forest classifiers for each vehicle and combining them through a reliable voting method. Additionally, each vehicle trains local classifiers using a random forest algorithm and leverages the collective knowledge of multiple vehicles. Gyawali et al. [15] proposed a novel mechanism to prevent internal attacks on vehicular systems. Their proposed mechanism combines AI techniques with a reputation

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mechanism to detect attacks and ensure the dependability of vehicles. The reputation mechanism is used to evaluate the trustworthiness of vehicles and their actions, while the AI component is used to analyze and interpret data to identify attacks or other issues that may affect the reliability of the vehicles. Tariq et al. [16] proposed a transfer learning-based approach for intrusion detection on a Controller Area Network (CAN). The authors employed a Convolutional LSTM network and leveraged pre-trained models from other domains to enhance the intrusion detection performance. Their approach showed promising results in accurately detecting intrusions at the CAN network, outperforming existing methods. Their proposed model is fine-tuned using one-shot transfer learning.

FL is a promising approach that can be used to improve the security and privacy of ITS. FL involves training local networks based on weights obtained from a central model. By training local networks on individual vehicles and aggregating their results at the central model, FL can enable vehicles to learn from each other and improve their performance while preserving their privacy. In this context, Lu et al. [17] developed a privacy-aware data protection mechanism for Vehicular cyber-physical systems (VCPS) that relies on FL. This approach involves (1) data transformation; and (2) collaborative data leakage detection. In the data transformation phase, data collected from individual vehicles is protected through techniques such as encryption and anonymization. The collaborative data leakage detection phase involves training models on the transformed data using FL, and then aggregating the results to detect any potential data leaks. This approach allows vehicles to share data and collaborate on tasks while still safeguarding the privacy of the data and protecting against potential threats. The scheme allows vehicles to locally train models while ensuring the privacy of their data. Kong et al. [18] presented an FL-based navigation scheme that utilizes a homomorphic encryption scheme along with the bounded Laplace mechanism to protect local model updates. Despite its advantages, FL is still at risk of certain types of attacks, including backdoor attacks. To protect against this type of attack, it is important to implement measures such as strong encryption and secure communication protocols, as well as to continually monitor and assess the performance and behavior of the models. Ensuring the reliability of participating nodes is crucial in order to prevent these types of attacks.

From another point of view, the integration of blockchain technology with FL to improve privacy protection in the ITS is of paramount importance. Lu et al. [19] proposed a blockchain-based approach to enhance secure data sharing in the Internet of Vehicles (IoV). Their proposed scheme used asynchronous federated learning, where data from multiple vehicles is aggregated and analyzed without sharing the raw data. By employing blockchain technology, the scheme ensures data integrity, transparency, and immutability. To efficiently select nodes, the framework employs a deep reinforcement learning (DRL)-adopted asynchronous FL approach. The proposed framework also includes learning models that enhance the reliability of data shared between a Roadside Unit (RSU) and vehicles. This is achieved through a two-step verification process, adding an extra layer of security and ensuring the integrity of the shared data. Liu et al. [20] proposed the integration of blockchain and federated learning to improve collaborative intrusion detection in vehicular edge computing. Their proposed work explores the use of blockchain technology to enhance the security and privacy of the data-sharing process. To secure the aggregation model, the authors proposed using blockchain to store and share the training models. Otoum et al. [21] focused on securing critical IoT infrastructures. The authors proposed a solution that combines blockchain and federated learning techniques to enhance the security and privacy of IoT systems. The use of blockchain technology ensures data integrity and immutability, while federated learning enables collaborative learning without exposing sensitive data to a central authority. Trusted end devices are included in the blockchain to ensure the accuracy of the consensus algorithm. Kang et al. [22] examined the issue of reliable federated learning in mobile networks. The authors proposed a solution that improves the reliability and efficiency of federated learning in the context of mobile networks. The study investigated various strategies to optimize the learning process and enhance the overall performance of federated learning systems.

After a careful review of the research literature, we identified several limitations. Some solutions were computationally expensive and were be suitable for resource-constrained environments. In addition, many of these solutions were trained on outdated datasets, which can lead to inaccurate and inefficient AI-based intrusion detection models when it comes to detecting zero-day attacks in vehicular networks. One of the main problems is the lack of up-to-date datasets containing a sufficient number of new attack records, as large institutions are often reluctant to share sensitive data with the research community due to privacy concerns. In light of these issues, we propose a secure collaborative Edge-based attack detection framework that uses FL and blockchain-based solutions to collaboratively and securely mitigate attacks in vehicular networks while preserving privacy and ensuring reliability.

III. System Model

In this section, we describe our system model; it outlines how the proposed framework ensures privacy-aware learning using FL and blockchain. In collaborative learning, as proposed by McMahan et al. [23], the objective is to find a global optimum solution through the collaboration and cooperation of multiple independent agents/systems without sharing their data. This can be thought of as a decentralized optimization problem, in which each agent/system has its own data and computational resources, and must work together to find a solution that is optimal for the group as a whole.

Federated optimization algorithms aim to coordinate the actions of these agents or systems in order to find an optimal solution in a distributed manner, without the need for a central authority or coordinator. Fig. 1 shows our proposed architecture which includes three tiers. In our three-tier federated learning system, the first tier consists of nodes in a vehicular network, such as vehicles equipped with sensors or communication devices. These nodes collect data and perform some initial processing, using local resources such as onboard

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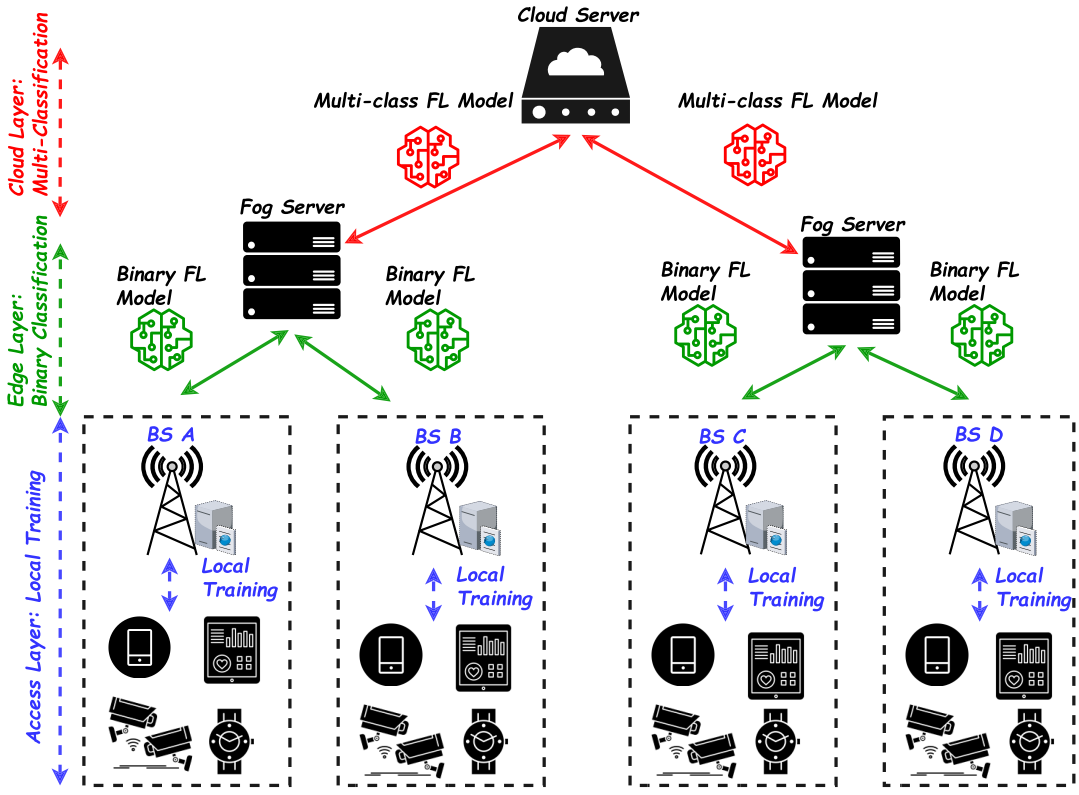


Fig. 1. System architecture.

processors or Edge computing nodes. The second tier, the Edge computing layer, consists of intermediate processing nodes located closer to the edge of the network, such as at traffic intersections or on-road infrastructure. These nodes perform more complex tasks, such as binary classification, using more powerful resources such as high-performance processors or GPUs. The third tier, the cloud tier, consists of centralized resources such as data centers or cloud servers, which are used to perform the most computationally intensive tasks such as multi-class classification. The three tiers work together to analyze and classify data collected from the vehicular network, using the resources available at each tier to perform the necessary processing and analysis. The goal is to find an optimal solution for the overall system, taking into account the constraints and capabilities of each tier. This approach allows the system to scale and adapt to changing conditions, such as changes in the data being collected or changes in the computational resources available at each tier. In our proposed architecture, the Edge layer serves as an intermediary between the first tier (consisting of nodes in a vehicular network) and the third tier (consisting of centralized resources such as data centers or cloud servers). The Edge layer aggregates the local model updates sent from each node in the vehicular network and uses them to perform real-time, low-latency attack detection using a binary classifier. Meanwhile, the cloud layer investigates the specific type of attack that has been detected, using a multi-class classifier to classify the attack into one of several categories (such as DDoS, worms, or fuzzers). This approach allows the system to quickly identify and classify attacks occurring in the vehicular network, using the resources available at each tier to perform the necessary processing and analysis.

In this process, the Edge organization is responsible for initiating and managing the collaborative learning effort for an ITS. This involves initializing the parameters of a shared FL learning model with an initial set of weights and selecting a group of N authenticated and reputable Edge-based participants to contribute to the learning process. The Edge organization may use various criteria to select these participants, such as their reputation or their ability to contribute high-quality data or computational resources to the learning process. Once the participants have been selected, the Edge organization can begin the federated learning process, coordinating the actions of the participants in order to find an optimal solution for the overall system (see Fig. 2). In the context of an ITS, the Edge organization is responsible for coordinating the collaborative learning process among a group of authenticated Edge-based participants. To do this, the organization first sends the shared model to each participant. Each participant then uses their own local ITS-based training data to perform local updates on the model, using techniques such as gradient descent or stochastic gradient descent. After the local updates have been completed, each Edge node encrypts its updated model and divides it into multiple shares. These encrypted model updates are then sent to a secure aggregator, which combines (aggregates) the updates from all of the participants and sends them back to

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the Edge organization. The Edge organization then decrypts the aggregated model and sends the resulting global model back to each Edge node, which can be used as the starting point for a new learning cycle. This process can be repeated iteratively, with the Edge organization coordinating the actions of the participants in order to find an optimal solution for the overall system [24], [25], [26], [27].

In this work, we aim to optimize the objective function of a non-convex neural network. Specifically, we define the optimization problem as follows:

$$\min_{w_r \in R^d} \varphi(w_r) \quad \text{where} \quad \varphi(w_r) = \frac{1}{N} \sum_{n=1}^{N} \varphi_n(w_r) \tag{1}$$

where N is the number of Edge participants.

The purpose of local learning is to find a set of parameters w_r that minimize the loss function, which can be expressed as follows:

$$\forall n, \qquad \varphi_n(w_r) = \frac{1}{J_n} \sum_{j_n=1}^{J_n} \varphi_{j_n}(w_r; x_{j_n}, y_{j_n}) \tag{2}$$

At the start of each round r in the collaborative learning process, each Edge node calculates the mean gradient on its local ITS-based data using the current model parameters w_r. This calculation is performed using a batch or local mini-batch of data and may involve one or several local epochs e. The process can be described as follows:

$$g_r^n = \nabla \varphi_n(W_r; b_r) \tag{3}$$

To perform local gradient descent on the shared FL model, each Edge collaborator will need to have access to the current model parameters and the local gradient that it calculated using its own local ITS-based data, as follows:

$$\forall n, \qquad W_r^n \leftarrow W_r - \eta \nabla \varphi_n(W_r; b_r) \tag{4}$$

IV. Subjective Logic

We are developing a novel blockchain-based reputation system for our data-sharing network, harnessing the power of subjective logic. Subjective logic is a probabilistic information fusion framework that enables the representation and combination of subjective beliefs using opinions. Our approach integrates the principles of subjective logic, allowing for the seamless representation and fusion of subjective beliefs and opinions on the blockchain. By incorporating this advanced methodology, our system accommodates a spectrum of positive, negative, and uncertain statements, enabling a comprehensive understanding of each participant's trustworthiness. With the incorporation of logical operators, we are establishing a sophisticated mechanism to correlate and analyze diverse opinions, thus fostering a robust and transparent evaluation of participants' interactions. This blockchain-based model ensures the seamless integration of subjective beliefs, paving the way for a secure and trustworthy environment within our data-sharing network.

A. Trusted Vehicle Opinions

Subjective logic serves as a valuable framework for managing uncertainties and subjective opinions within multi-agent systems. In this context, the local opinion vector, denoted as $O_i^j = \{b_i, d_i, u_i, a\}$, plays a pivotal role in capturing the nuanced dynamics of trust and belief between individual vehicles. Here, b_i signifies the confidence or belief that vehicle K_i places in the information relayed by vehicle K_j, while d_i reflects the corresponding level of doubt or disbelief. Complementing these, the parameter u_i denotes the degree of uncertainty that vehicle K_i associates with the transmitted data. Together, these components provide a comprehensive representation of the intricate nature of trust and belief within the vehicular network. Finally, the parameter a assumes significance as the base rate, influencing the calculation of the expectation value of the probability of opinion, denoted as $E(O_i) = b_i + a u_i$. By incorporating these elements, the local opinion vector facilitates a structured and systematic approach to managing trust relationships, enabling vehicles to make well-informed decisions in the face of uncertainties. This is calculated as follows:

$$\begin{cases} b_{i\to j} = \dfrac{\alpha_{i\to j}}{\alpha_{i\to j} + \beta_{i\to j}} \\ d_{i\to j} = \dfrac{\beta_{i\to j} + 2a - 2}{\alpha_{i\to j} + \beta_{i\to j}} \\ u_{i\to j} = \dfrac{2}{\alpha_{i\to j} + \beta_{i\to j}} \\ a = \text{base rate} \end{cases} \tag{5}$$

The local opinion vector u_i^j contains α positive events and β negative events with uncertainty represented by a. The extent of uncertainty within the vector is intricately tied to the efficacy of communication established between vehicles i and j. This communication quality, represented as $1-b_{i\to j}-d_{i\to j}$, serves as a metric for evaluating the likelihood of successful data packet transmission during the exchange of information for data sharing requests.

Subjective logic encompasses various types of opinions, including hyper opinions, multinomial opinions, and binomial opinions. Within the scope of this article, we concentrate on uncertain binomial opinions (UBO) concerning binary events, particularly those associated with valid or invalid data transmission. This selection is informed by the inherent uncertainty prevalent in updates or data transmissions among vehicles within the transportation system, always bounded from zero on-wards. As a result, the local reputation $T_{i\to j}^{local}$ can be defined as follows:

$$T_{i\to j}^{local} = b_{i\to j}^{local} + \gamma u_{i\to j}^{local} \tag{6}$$

where $\gamma \in [0, 1]$ denotes the level of uncertainty that impacts the reputation of vehicles.

B. Multi-Weighted Local Opinions

Multi-weight subjective logic is an evolution of traditional SL that involves weighting operations. In our framework, different weights are used to formulate local opinions. In this

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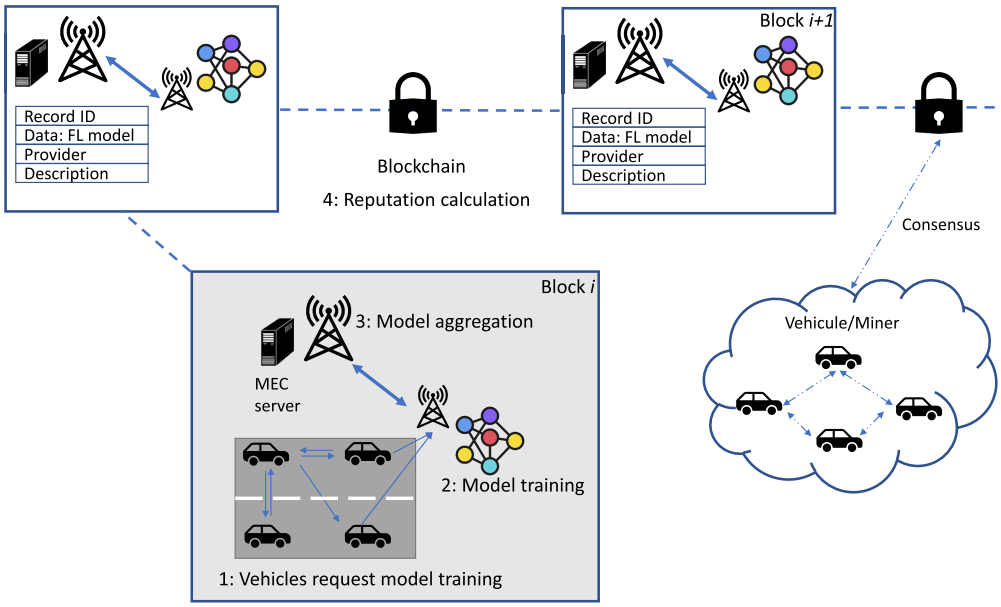


Fig. 2. System process.

context, the following factors are taken into consideration when calculating the reputation of a vehicle:

1) Vehicle Interaction: A higher frequency of interaction indicates that the vehicle data requestor (k_i) has more pre-existing information about the vehicle data provider (k_j), thereby facilitating a more precise and dependable calculation of reputation. Vehicles can either interact honestly or dishonestly. Honest interactions increase the trust of the vehicle and result in an increase in reward, while dishonest interactions have the opposite effect. We can define a vehicle interaction VI as the ratio of the number of interactions between the data requestor and the data provider during a time window, as follows: possesses more pre-existing information about the data provider (k_j), thereby facilitating a more precise and dependable calculation of reputation.

$$I_{i\to j} = V\frac{\alpha_{i\to j}+\beta_{i\to j}}{\sum_{k\in V}\alpha_{i\to k}+\beta_{i\to k}} \tag{7}$$

2) Event Updates: In our framework, the trustworthiness and reliability of a vehicle can change over time. The reputation of a data provider (k_j) from the perspective of a data requestor (k_i) is affected by both recent and past events, with the former having a larger impact on the local opinion. Typically, recent events are considered within a one-week timeframe, and negative events carry greater weight in shaping local opinions than positive ones. This facilitates the generation of a reputation calculation that is more precise and dependable.
Event updates reflect the effect of time on reputation, in this context, the weight of positive events is represented by $w1^{t_{curr}}$, while the weight of negative events is represented by $w2^{t_{curr}}$. It is stated that $w1+w2=1$, and that $w1 \le w2$, implying that negative events are given more weight than positive events. Additionally, σ and τ where $\sigma+\tau=1$, represents the weights of recent event and the weights of past events, respectively, as follows:

$$\begin{cases} \alpha_i = w1^{t_{curr}}(\sigma\alpha_i^{t_{curr}}+\tau\alpha_i^{t_{pas}}) \\ \beta_i = w2^{t_{curr}}(\sigma\beta_i^{t_{curr}}+\tau\beta_i^{t_{pas}}) \end{cases} \tag{8}$$

where $\alpha_i^{t_{curr}}$ and $\alpha_i^{t_{pas}}$ represent the number of current positive events and past positive events, respectively. In this context, a vehicle interaction can be formulated as follows:

$$I_{i\to j} = V\frac{w1^{t_{curr}}(\sigma\alpha_i^{t_{curr}}+\tau\alpha_i^{t_{pas}})+w2^{t_{curr}}(\sigma\beta_i^{t_{curr}}+\tau\beta_i^{t_{pas}})}{\sum_{k\in V}\alpha_{i\to k}+\beta_{i\to k}} \tag{9}$$

3) Trajectory Similarity: In transportation systems, data collected by vehicles is considered to be relevant only to the vehicles that collected it, and is limited in scope to the specific location where it was collected. To improve the relevance of the data and enable location awareness, the trajectory similarity of the vehicles is taken into consideration when calculating reputation during data sharing among vehicles. A high trajectory similarity indicates that the data being shared from the data provider is more relevant and likely to be of high quality, accuracy, and reliability. This consideration of trajectory similarity is intended to help ensure that data sharing among vehicles results in the most useful and relevant information being shared.

Our framework introduces a trajectory similarity measure that builds upon a modified version of the Hausdorff distance. This measure is designed to capture the similarity between two trajectories and can be used to compare and analyze the movements of objects or individuals over time. Given a universal set T and Hansdroff distance $d : TxT \to \mathbb{R}$ between

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$V \in T$ and $W \in T$, as follows:

$$h(V, W) = \max\{\sup_{v \in W} d(v, W), \sup_{w \in V} d(V, w)\} \quad (10)$$

where $h(V, W) \neq h(W, V)$. As an extension we can define $h(V, W) = \max\{h(V, W), h(W, V)\}$.

The Hausdorff distance is a measure of similarity between sets of points, but it has limitations when used to compare sets with outlier points or time series data. It is affected by the presence of outlier points and is not designed to consider the order of points in a series. In order to improve the accuracy and reliability of the matching process, we take two actions: rejecting a certain proportion of the worst matches identified under the maximum specified in the equation, and restricting the set over which the minimum is calculated to a subset of the second set. These actions aim to reduce the impact of poor-quality matches and focus the matching process on a specific subset of the second set, improving the overall performance of the process, as follows:

$$h_{\alpha,N,C}(V, W) = ord^{\alpha}_{v \in V}\{\min_{w \in N_W(C_v, W(v))} d(v, w)\} \quad (11)$$

where $C_{V,W} : V \rightarrow W$ maps each point $v \in V$ to a corresponding point $w \in W$ and $N_W : W \rightarrow V(W)$ maps each point w, which represents the neighborhood of that point v. The functions N_W and $C_{V,W}$ work together to create a structure within which the matching process is performed. The operator $ord^{\alpha} f(x)$ refers to the value in the set $f(x)$, i.e., the image of the set x under the function f, that is grater than α times the number of values in $f(x)$, Here:

$$ord^{0}_{v \in V}\{f(v)\} = \min_{v \in V} f(v) \quad (12)$$

$$ord^{0.5}_{v \in V}\{f(v)\} = \check{f(v)} \quad (13)$$

$$ord^{1}_{v \in V}\{f(v)\} = \max_{v \in V} f(v) \quad (14)$$

Our aim is to identify the spatial patterns within the trajectories. To achieve this, we've opted for arc-length parameterization. We define the length of a specific trajectory t as follows:

$$|T| = \sum_{i \in V} \|t_i - t_{i-1}\|^2 \quad (15)$$

Additionally, we define the subsequent formed by the first k vehicle/points as $T_k = \{t_i\}_{i=1}^{k}$. We also define the relative position of the nth vehicle/point in T as:

$$\pi_t(t_n) = \frac{|T_n|}{|T|} \quad (16)$$

To express the correspondence structure $C_{V,W}$, we define the reverse transformation as follows:

$$r_T(x) = T(arg\min_{n \leq V} |a - \frac{|T_n|}{|T|}|) \quad (17)$$

$$C_{V,W}(v) = W(r_W(\pi_v(v))) \quad (18)$$

The neighborhood structure N_W is defined as:

$$N_W(w_0) = \{w \in Q \| \pi_W(w_0) - \pi_W(w)| \leq w/2\} \quad (19)$$

where w is the parameter relative to the size of the neighborhood.

The Hausdorff distance is now represented by α, w, where α and w are the specified parameters. It should be understood that the arc-length parameterization is used for the correspondence in N_W and $C_{V,W}$, and w is used as the neighborhood size. We can express the similarity as follows:

$$sim(v, w) = \exp\{-\frac{d^2(v, w)}{2\alpha}\} \quad (20)$$

where $d(v, w)$ is the squared distance between v and w. The overall weight of reputation for local opinions is:

$$\sigma_{i \rightarrow j} = \gamma_1 I_{i \rightarrow j} + \gamma_2 sim(i, j) \quad (21)$$

where $\gamma_1 + \gamma_2 = 1$, and $0 \leq \gamma_1 \leq 1$ and $0 \leq \gamma_2 \leq 1$.

C. Recommended Opinions

In the context of blockchain networks, the consideration of reputation opinions is crucial. Miners, who play a significant role in maintaining the integrity of the blockchain, rely on various sources of information to make informed decisions. Apart from local reputation opinions, they also delve into the blockchain to gather direct reputation opinions of selected workers, updated by other nodes. This process enables them to obtain recommended reputation opinions. Let's assume that miner m receives a set of recommended opinions about worker $w^{rec}_{i \rightarrow j} := b^{rec}_{i \rightarrow j}, d^{rec}_{i \rightarrow j}, u^{rec}_{i \rightarrow j}$, as follows:

$$w_r = \frac{b_{i \rightarrow r}(b^{local}_{r \rightarrow w} + d^{local}_{r \rightarrow w})}{\sum_{r \in X}(b_{r \rightarrow w} + d^{local}_{r \rightarrow w})} \quad (22)$$

where $r \in X$ is the recommender and $d_{r \rightarrow w} + d_{r \rightarrow w}$ is the familiarity value between recommender r and worker w and. This value represents the similarity or relatedness between the two recommenders or items, with a higher value indicating a higher level of familiarity. In our framework, the set of blockchain workers $\Phi(W)$ provide a set of recommendation $\Phi(r)$, we can express:

$$b_{i \rightarrow r} = \frac{|\Phi(W_i) \cap \Phi(r)|}{|\Phi(W_i) \cup \Phi(r)|} \quad (23)$$

Finally,the recommended opinion is a composite of subjective opinions from various sources, such as neighboring vehicles. These opinions are integrated and weighted to produce a single opinion, as:

$$\begin{cases} b^{rec}_{i \rightarrow w} = \sum_{r \in X} w_r b^{rec}_{i \rightarrow w} \\ d^{rec}_{i \rightarrow w} = \sum_{r \in X} w_r d^{rec}_{i \rightarrow w} \\ u^{rec}_{i \rightarrow w} = \sum_{r \in X} w_r u^{rec}_{i \rightarrow w} \end{cases} \quad (24)$$

D. Integrating Local Opinions With Recommended Opinions

After acquiring shared data from multiple data providers, a data requestor might construct a subjective perception of each provider, influenced by their past interactions. To ensure the integrity of the process, it is essential to integrate these individual viewpoints while shaping the ultimate, unified opinion. The combined reputation opinion of v_i from worker w is

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denoted as $R^{com}_{i\to w} = b^{com}_{i\to w}, d^{com}_{i\to w}, u^{com}_{i\to w}$ and can be defined as follows:

$$\begin{cases} b^{com}_{i\to w} = \dfrac{b^{local}_{i\to w}u^{rec}_{i\to w} + b^{rec}_{i\to w}u^{local}_{i\to w}}{u^{local}_{i\to w} + u^{rec}_{i\to w} - u^{rec}_{i\to w}u^{local}_{i\to w}} \\ d^{com}_{i\to w} = \dfrac{d^{local}_{i\to w}u^{com}_{i\to w} + d^{rec}_{i\to w}u^{local}_{i\to w}}{u^{local}_{i\to w} + u^{rec}_{i\to w} - u^{rec}_{i\to w}u^{local}_{i\to w}} \\ u^{com}_{i\to w} = \dfrac{u^{local}_{i\to w}u^{rec}_{i\to w}}{u^{local}_{i\to w} + u^{rec}_{i\to w} - u^{rec}_{i\to w}u^{local}_{i\to w}} \end{cases} \tag{25}$$

Similarly, the composite reputation is:

$$T^{com}_{i\to j} = b^{com}_{i\to j} + \gamma u^{com}_{i\to j} \tag{26}$$

V. Implementation

In this section, we present the steps taken to implement and evaluate our proposed framework. We begin by outlining the experimental environment in which the framework was tested. Next, we present the experimental results and provide an analysis of the performance of our proposed framework. This includes well-known metrics to measure the effectiveness of AI models. Finally, we evaluate the effectiveness of the framework based on the experimental results. This includes a summary of the key findings from the experiments, as well as any recommendations or next steps that we believe would be beneficial for improving the framework in the future.

A. Experimental Results

We have implemented our proposed framework using Pysyft [28] to easily build and train deep learning models in a federated setting, leveraging the powerful capabilities of PyTorch to build and train our models. To simulate the ITS-based environment test, we use Mininet [29], a virtual environment for testing and simulating the behavior of vehicular networks. Mininet allows users to create a virtual networks, consisting of virtual hosts, switches, and links, and to test how the network behaves under different conditions. Mininet can be used to simulate a variety of network scenarios, including those related to vehicular networks. In our tested scenario, Mininet was used to create a virtual environment that includes virtual vehicles equipped with sensors or communication devices, as well as virtual infrastructures such as traffic lights or road signs.

In order to evaluate the efficacy and practicality of our proposed framework, we carried out experiments utilizing the UNSW-NB15 dataset, specifically focusing on attacks within ITS. This dataset, renowned for its substantial volume of data samples, served as an ideal resource for evaluating the framework's performance; it includes 100 GB of attack traffic, such as backdoors, exploits, and shellcodes, which are common types of attacks in ITS. By using the UNSW-NB15 dataset to evaluate our framework, we were able to assess its performance in detecting different types of attacks and identify any potential issues in ITS-based networks. This dataset provided a realistic and representative testbed for evaluating the effectiveness of our proposed framework in an ITS setting.

TABLE I
UNSW-NB15 Labels

Label	Multi-class	Binary
Normal	0	0
Fuzzers	1	1
Analysis	2	1
Backdoors	3	1
DoS	4	1
Exploits	5	1
Generic	6	1
Reconnaissance	7	1
Shellcode	8	1
Worms	9	1

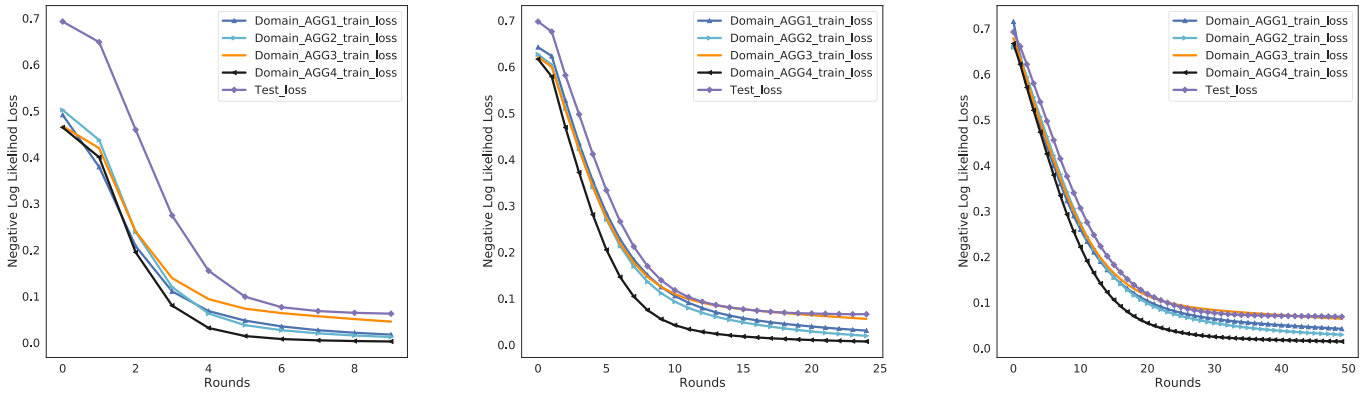
In this study, we have defined two different scenarios to test the effectiveness of our proposed framework: one for multi-class classification and one for binary classification. In the multi-class classification scenario, the model must predict the correct class label from a set of multiple class labels. In the binary classification scenario, the model must predict between two class labels, typically represented as 0 (Normal) and 1 (Attack). We divide the UNSW-NB15 dataset in order to give each participant a subset of the data and also to deal with issues with non-i.i.d (as in "non-independent and identically distributed data"). To evaluate the performance of our framework, we tested a range of combinations of the number of training rounds (varying from 10 to 50) and local epochs (varying from 1 to 5). We used four participants (A, B, C, and D, as shown in Fig. 1) in each scenario, each working with their local data. To begin, we randomly initialized the parameters of the global model and sent it to each participant. Each participant then trained the model using their local data, and the global model was tested on the test set. This process allowed us to evaluate the performance of the model across all of the participant's data, including in the case of an imbalanced dataset.

Fig. 3 shows the learning curves of the tested models in different scenarios, as represented by the negative log-likelihood loss values, the x-axis represents the number of training iterations or epochs, while the y-axis represents the negative log loss. The curve shows the negative log loss decreasing over time as the model is trained. As the FL model becomes more accurate, the negative log loss decreases, eventually reaching a minimum value where the FL model is most accurate. This shows that the trained models used to detect zero-day attacks in the ITS setting are able to learn from each other without sharing their local data. This is achieved through the use of our proposed framework, in a decentralized privacy-aware manner.

VI. Performance Evaluation

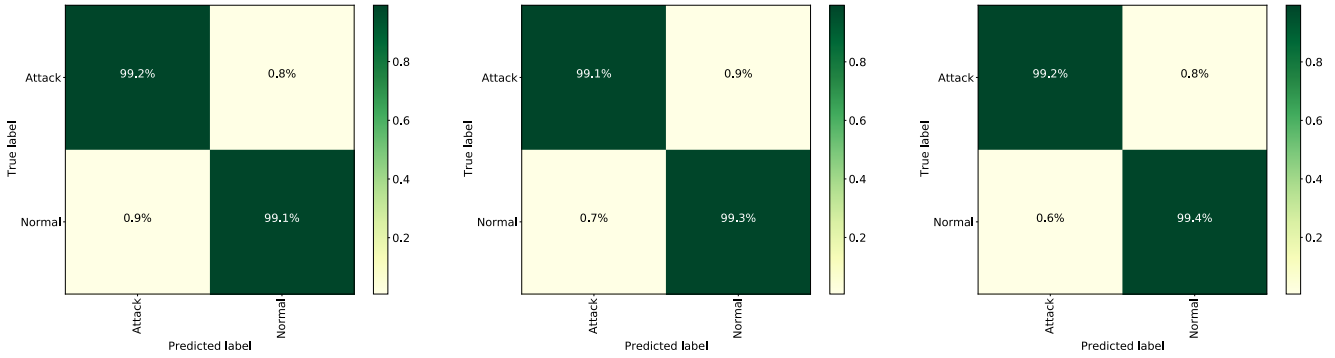
To evaluate the performance of our proposed framework, we used a variety of metrics, including accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic (ROC) curve (AUC). These metrics allow us to measure the effectiveness of the framework in different ways, including its ability to classify instances accurately, identify true positives and negatives, and distinguish between different classes. The area under the ROC curve (AUC): This metric measures the performance of the model in distinguishing

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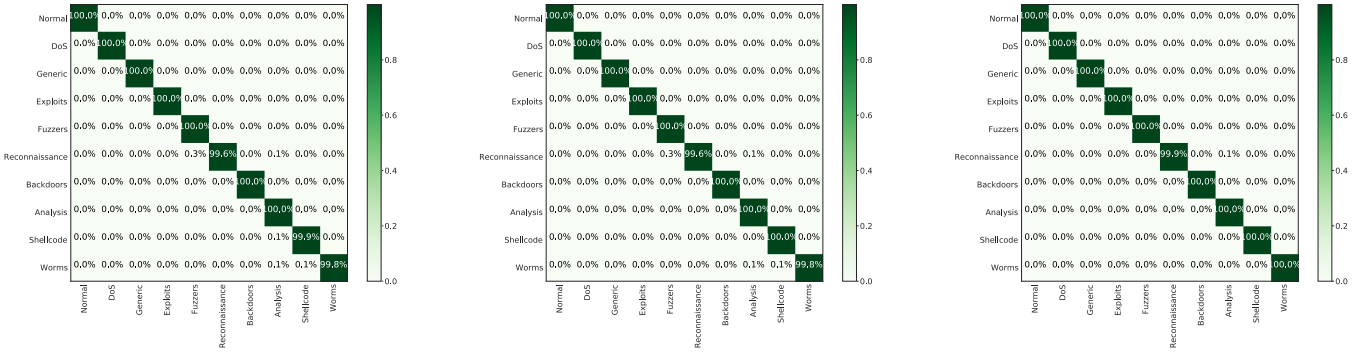
(a) (b) (c)

Fig. 3. Learning curves for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.



(a) (b) (c)

Fig. 4. Confusion matrices for Binary classification for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.



(a) (b) (c)

Fig. 5. Confusion matrices for Multi-class Classification for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

between positive and negative instances, calculated as the area under the ROC curve. A model with a higher AUC score is considered to be more effective at distinguishing between positive and negative instances. These metrics are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{27}$$

$$Precision = \frac{TP}{TP + FP} \tag{28}$$

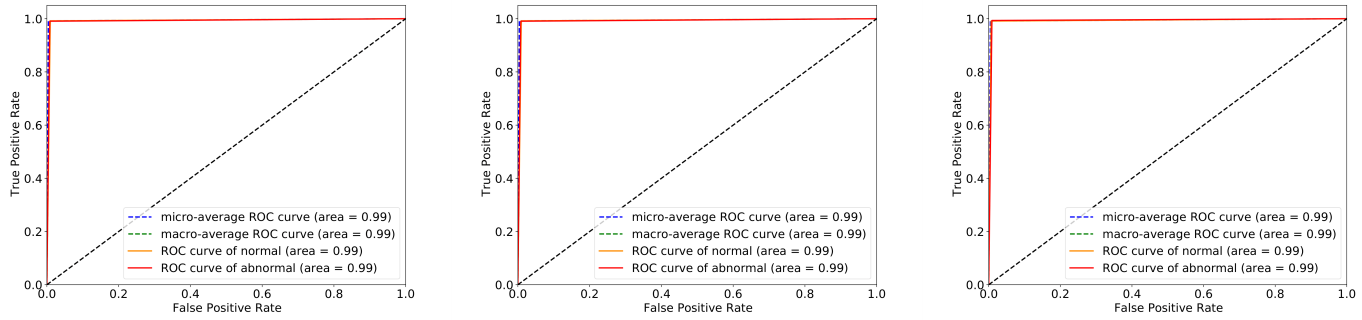
$$Recall = DR = TPR = \frac{TP}{TP + FN} \tag{29}$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{30}$$

$$FPR = \frac{FP}{TN + FP} \tag{31}$$

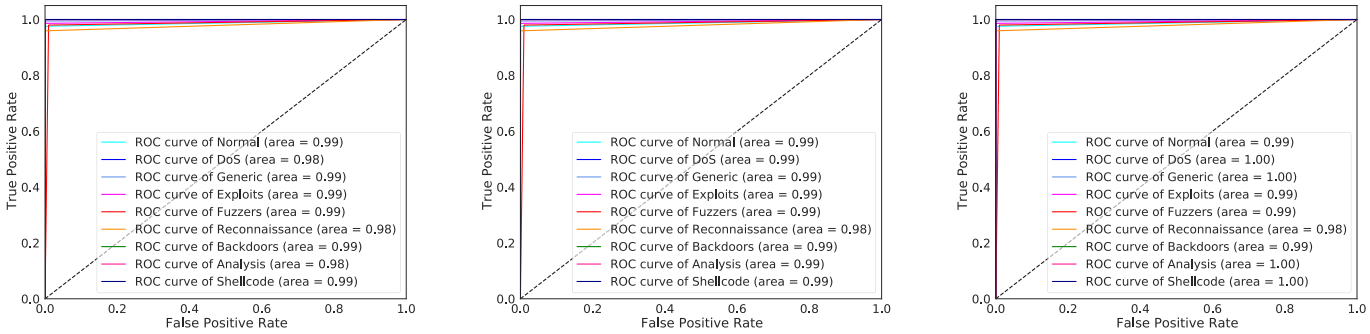
Figs. 4 and 5 show the confusion matrices on the UNSW-NB15 test dataset for binary (Edge-based training) and multi-class (cloud-based training) classification, respectively. These obtained results show that our proposed framework has achieved high performance in detecting the new attacks in ITS, as indicated by the high values of accuracy, precision, recall, and F1 score. The obtained accuracy, precision, recall,

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(a) (b) (c)

Fig. 6. ROC Curves for Binary classification for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.



(a) (b) (c)

Fig. 7. ROC Curves for Multi-class Classification for (a) 10 rounds; (b) 25 rounds; and (c) 50 rounds.

and F1-score of 99%, respectively, in the Edge-based training step (binary classification) and the cloud-based step (multi-class classification) shows that the proposed framework is able to correctly classify a large proportion of the ITS-based attacks of the UNSW-NB15 dataset. Thus, the results obtained in Figs. 4 and 5 show that the proposed framework is highly effective at detecting ITS-based attacks with high accuracy and privacy.

Figs. 6 and 7 show the confusion matrices on the UNSW-NB15 test dataset for binary (Edge-based training) and multi-class (cloud-based training) classification, respectively. In both scenarios, our proposed framework achieves an AUC of 99%. These results demonstrate that our proposed framework is highly effective at detecting ITS-based attacks with high AUC in both scenarios while preserving participants' data privacy.

A. Comparative Analysis

In this section, we evaluate the performance of our proposed framework with centralized AI-based schemes, using the UNSW-NB15 dataset. To compare the performance of these approaches, we use various evaluation metrics, including accuracy, precision, recall, and F1-score.

In this study, we compare the performance of our proposed framework with several recent centralized AI-based schemes, including: SVM [30], StackEns [31], MultiStacking [32], HybridML [33], LSTM [34], OnlineEns [35], TSE [36], StackExtra [37], and IDS-OGM [38]. Table II presents a comparison of the accuracy, precision, recall, and F1-score of our proposed framework with centralized AI-based schemes. The results show that our proposed framework has the highest accuracy of 99% and the highest F1-score of 99%. These results demonstrate the effectiveness of our proposed framework in detecting attacks while preserving the privacy of the participants.

TABLE II
COMPARATIVE ANALYSIS

Methods	Accuracy	Precision	Recall	F1
SVM	0.98	0.9	0.97	0.93
StackEns	0.94	N/A	N/A	N/A
MultiStacking	0.89	N/A	N/A	N/A
HybridML	0.95	0.92	0.92	0.92
LSTM	0.92	N/A	N/A	N/A
TSE	0.85	0.87	0.88	N/A
StackExtra	0.96	N/A	N/A	N/A
IDS-OGM	0.95	0.94	N/A	N/A
Our Proposed Framework	**0.99**	**0.99**	**0.99**	**0.99**

VII. CONCLUSION

In this paper, we introduced a novel framework for securing ITS against emerging threats using FL and blockchain technology. Our proposed framework includes a distributed Edge-based architecture that allows multiple Edge nodes to collaborate in securing the ITS while preserving their privacy. Then, we have introduced a decentralized and secure reputation system based on blockchain technology to maintain the reliability and trustworthiness of the FL process within the ITS. This system manages reputation data for individual nodes (such as vehicles), guaranteeing the integrity of the FL

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training process. Evaluation of the proposed framework using the UNSW-NB15 dataset, which contains well-known attacks and malware, showed that our proposed framework achieved high accuracy and F1 score of 99%, respectively, in detecting ITS-related threats while preserving privacy and reliability. These results demonstrate the effectiveness of our proposed framework in securing ITS. For future work, we intend to develop AI models that offer explanations, thereby increasing the transparency of the FL training process and facilitating the identification and prevention of backdoor attacks. Furthermore, we aim to expand and enhance the scalability of the reputation process. Lastly, we will extend the application of our framework to other datasets to ensure comprehensive testing and validation.

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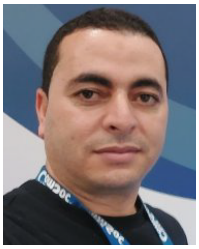
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